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Investigating Clouds and Snowfall in
Antarctica: Insight from Lidar and
Radar Data at Scott Base

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DECLARATION

I certify that content of this report was mostly my own work.

My supervisor helped by proofreading the report and provided usual supervisory support with respect to the use of the ALCF software package.

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ABSTRACT

Antarctica remains one of the least understood regions due to its remoteness and extreme climate, particularly in relation to cloud and snowfall processes. Climate models often struggle to simulate these aspects accurately, leading to biases, especially in cloud representation. Since clouds and snowfall regulate Earth's radiative and surface mass balances, understanding them is critical. This study utilized Vaisala CL51 lidar and MRR2 radar to investigate cloud properties and snowfall over Antarctica. Data were reprocessed using the Automatic Lidar and Ceilometer Framework (ALCF), and radar data were used to derive reflectivity, Doppler velocity, and spectral width, using method proposed in Maahn and Kollias (2012). We identified supercooled liquid water clouds (SLWC) at 4 km altitude, with lidar backscatter values exceeding $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$. Seasonal variations in clouds were observed, with higher levels during summer and autumn at 3 km, which align with snow accumulation patterns showing greater build-up during these seasons. While we found evidence of SLWC contributing to snowfall in one event, further analysis is required to better understand the relationship between clouds and snowfall using combined lidar and radar data.

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1 Introduction

Clouds and snowfall are crucial to the global radiative and water balance, as they can modify the planetary albedo, influencing the amount of solar energy that is either reflected or absorbed by the Earth. In addition, clouds play a significant role in regulating longwave radiation, either by trapping heat or allowing more of it to escape into space. As the primary source of precipitation, clouds contribute to snow formation, which is the largest input for the surface mass balance of the Antarctic ice sheet (Wiener et al., 2024). Despite their importance, supercooled liquid water clouds (SLWC), in particular, are often underrepresented in climate models. This underrepresentation results in a low bias in simulated reflected shortwave radiation at latitudes higher than 55°S , largely due to poor phase representation in the dominant SLWC regime (Standford et al., 2023). Supercooled liquid water clouds, which contain water droplets at temperatures below 0°C but above the homogeneous nucleation temperature of around -40°C , are vital to accurately modelling precipitation. Ice formation in clouds occurs through heterogeneous nucleation, where supercooled droplets collide with ice-nucleating particles (INPs) like dust and aerosols (Whitehead et al., 2024). The unique properties of SLWC can influence snowfall, contributing to both light and heavy snow events depending on atmospheric conditions. Understanding SLWC's role in precipitation can improve climate models, enhance weather forecasting, and inform climate policies and adaptation strategies.

Clouds form through the process of condensation, where warm, moist air cools and transforms into tiny water droplets, leading to cloud formation. The properties of cloud water content vary by region, with polar areas, such as Antarctica, containing more ice and supercooled liquid water (SLW) clouds, while regions near the tropics tend to have more warm liquid clouds. In Antarctica, the colder temperatures favour the formation of ice and SLW clouds, while mixed-phase clouds—containing both solid ice and liquid water—are frequently observed near coastal areas. These clouds often feature larger ice crystals and water droplets (Ricaud et al., 2024). The coexistence of ice crystals and liquid water in clouds impacts precipitation formation through the Wegener-Bergeron-Findeisen process, where ice crystals grow by absorbing moisture from nearby supercooled droplets, eventually falling as snow. This process underscores the importance of understanding how different cloud compositions influence the amount of precipitation.

In this study, we will employ ground-based lidar and radar methods to gain a better understanding of clouds and snowfall at Scott Base, Antarctica. The region's severe weather conditions and its tropospheric column, which extends to altitudes of 8 to 10 kilometers, make it a challenging yet critical environment for atmospheric research. Due to the cold, dense air in polar regions like Antarctica, the troposphere is thinner, which affects cloud formation and weather patterns. Many previous studies have investigated clouds and snowfall in polar regions, often using separate methods for each. For instance, Guyot et al. (2022) and Aubry et al. (2024) proposed different lidar-radar techniques to study cloud content, with Guyot et al. (2022) focusing on identifying supercooled liquid water content (SLWC). However, both studies did not delve into the connection between SLWC and precipitation. On the other hand, radar studies by Souverijns et al. (2017), Scarchilli et al. (2020), and Gorodetskaya et al. (2015) focused on snowfall observation, effectively deriving snowfall rates using radar technology. What has been missing in previous research is a comprehensive approach that combines both lidar and radar to

simultaneously study cloud properties and their role in precipitation.

We also aim to observe how SLWC may contribute to heavy snowfall events. By integrating data from both instruments, we hope to improve our understanding of the role SLWC plays in producing heavy snowfall and enhance detection accuracy in Antarctica's unique environment. Furthermore, this research seeks to explore under what conditions SLWC or other cloud types are responsible for significant snowfall, with the ultimate goal of improving climate models and weather forecasting for extreme events. This work will offer valuable insights into cloud-precipitation interactions in polar regions and provide a strong foundation for future studies in this field.

Lidar is an active remote sensing instrument that emits pulses of laser light into the atmosphere and measures the backscattered radiation from atmospheric particles such as liquid droplets, ice crystals, aerosols, and other constituents. It has become one of the most widely used tools for studying cloud properties through both ground-based and satellite-based measurements. However, many studies, such as McErlach et al. (2021), have shown that satellite instruments often struggle to detect low-altitude clouds due to signal attenuation caused by higher-altitude clouds, leading to inaccuracies in measuring low-level cloud occurrences. To obtain precise measurements of the lower troposphere, this study employs ground-based remote sensing techniques, specifically a Vaisala CL51 ceilometer. The ceilometer enables high-resolution observations of clouds in the troposphere, both spatially and temporally. This instrument is highly effective at detecting cloud base height and vertical cloud distribution. Lidar instruments operating at wavelengths between 355 and 1064 nm, are sensitive to hydrometeor concentrations and can accurately detect cloud properties (Aubry et al., 2024). Advanced methods, including machine learning models trained on ceilometer data, have been developed to detect supercooled liquid water (SLWC) with accuracies as high as 91% (Whitehead et al., 2024, Guyot et al., 2022). However, lidar is not well-suited for measuring precipitation, as its signal becomes heavily attenuated when interacting with large raindrops or snowflakes, reducing its ability to accurately capture precipitation rates.

Radar technology is widely used for measuring precipitation, particularly snowfall, due to its ability to detect snowflakes through their reflectivity, providing detailed insights into the intensity and structure of snowfall. Numerous studies have combined radar and lidar data to enhance understanding of both cloud and precipitation processes. For example, Standford et al. (2023) utilized radar and lidar datasets to estimate precipitation occurrence from liquid cloud bases (LCB). In this study, we adopted the method proposed by Maahn and Kollias (2012), utilizing data from the Micro Rain Radar 2 (MRR2), a profiling Doppler radar, to derive effective radar reflectivity (Ze), Doppler velocity, and spectral width from the radar's raw spectrum. This data is critical for assessing snow accumulation and the direction of precipitation relative to the instrument. Additionally, we analysed snowfall rates using a power law relationship,

$$Ze = aSr^b \quad (1)$$

Where the coefficients a and b were determined based on local conditions and snow types. Souverijns et al. (2017) found that the prefactor is sensitive to the median diameter of snowflakes, while the exponent b remains unaffected by this parameter. In this study, we used Ze - Sr relationships derived by Kulie and Bennartz (2009) and Matrosov (2007) to

evaluate snowfall rates and accumulation, offering valuable insights into snow processes. The Ze-Sr relations can be found in Table 1.

We have conducted the analysis primarily by examining variations in cloud cover and precipitation over specific time periods. We also consider how snowfall varies under different synoptic weather patterns. This approach is crucial for identifying which weather conditions are more likely to produce higher or lower snowfall rates. Studies such as McDonald et al. (2024) have demonstrated that classifying wind patterns using a Self-Organizing Map (SOM) approach can reveal variations in cloud occurrence across different reanalysis products and ceilometer data. Similarly, Jolly et al. (2018) applied a k-means clustering method to 33 years of surface wind data from the ERA5 reanalysis product, showing significant variations in cloud occurrence based on synoptic weather types. These findings highlight the importance of understanding how changes in weather patterns can influence cloud formation in the troposphere, potentially leading to shifts in snowfall patterns.

By applying synoptic weather pattern classification methods to snowfall data, researchers can gain valuable insights into how different weather regimes impact snowfall variability in regions like Antarctica. This study will adopt the SOM methodology as suggested by McDonald (2016), which is a clustering technique often used to derive weather regimes. SOMs are artificial neural networks designed for dimensionality reduction by clustering, adjusting reference vectors iteratively based on input data differences (Kohonen, 1990). Previous studies have shown that this classification method produces clusters that align geographically and seasonally with results from k-means clustering (McDonald, 2016). Moreover, weather states derived using this method reflect different pressure systems, influenced by wind circulation patterns. In the Southern Hemisphere, cyclonic circulations are typically associated with low-pressure systems, while anticyclonic circulations correlate with high-pressure systems (Marshall and Plumb, 2008). These pressure systems affect the amount of vapor transported into the troposphere, influencing cloud formation and subsequent snowfall.

2 Methodology and datasets

2.1 Vaisala CL51 ceilometer

For this campaign, we deployed the Vaisala CL51 ceilometer, a sophisticated instrument designed for precise cloud observations. Widely used in atmospheric studies, the CL51 excels at measuring Cloud Base Height (CBH), a crucial parameter for understanding cloud formation, structure, and atmospheric behavior. In addition to CBH, this instrument provides detailed vertical profiles of the attenuated volume backscattering coefficient, capturing important information about how atmospheric particles, such as aerosols and hydrometeors, interact with light. As noted in studies by Kuma et al. (2021), this capability is crucial for characterizing cloud properties and improving the accuracy of weather and climate models.

The CL51 was deployed at Scott Base, Antarctica, and operated continuously for three years, from 2021 to 2023. Our analysis primarily focuses on atmospheric data collected in 2022, within one of the most extreme environments on Earth. Operating at a near-infrared

wavelength of 910 nm, the CL51 can detect atmospheric features up to 15 km in altitude, ensuring observations of both low- and high-altitude clouds. Its high vertical resolution of 10 meters provides precise measurements, while a time resolution of 6 seconds captures rapid changes in atmospheric conditions.

After collecting the raw data, we convert it into NetCDF format using the `cl2nc` tool (Kuma et al., 2021), a necessary step to facilitate compatibility with widely used scientific processing tools. The resulting NetCDF files will be further processed using the Automatic Lidar and Ceilometer Framework (ALCF), which automates the interpretation of lidar and ceilometer data. This approach will enable a deeper analysis of cloud structures, atmospheric layering, and particle distributions, yielding valuable insights essential to achieving the campaign’s scientific objectives.

2.2 Metek Micro Rain Radar 2 (MRR2)

The Metek Micro Rain Radar 2 (MRR2) is a powerful radar frequently used in precipitation studies, with a particular focus on understanding rain. Although originally designed to detect liquid rainfall, its potential for observing snowfall has been well-established through research by Matrosov (2008) and further supported by Berne and Krajewski (2013). Kneifel et al. (2011) later evaluated this capability for more affordable radar systems like the MRR2, confirming its effectiveness. Operating as a frequency-modulated continuous-wave Doppler radar, the MRR2 transmits at 24 GHz (corresponding to a wavelength of 1.24 cm) via a 50mW solid-state transmitter, with its signals concentrated by a 60 cm offset parabolic antenna. For this study, the radar was positioned alongside the Vaisala CL51 ceilometer and operated continuously at the same location over three years from 2021 to 2023. To extract meaningful information from the raw data, we apply an advanced processing technique developed by Maahn and Kollias (2012), which optimizes the MRR2’s detection of solid precipitation. This method enhances the radar’s ability to provide accurate measurements of effective reflectivity (Z_e , expressed in $\frac{\text{mm}^6}{\text{m}^3}$ or dBz), mean Doppler velocity ms^{-1} , and velocity spectral width ms^{-1} , the latter offering valuable insights into atmospheric turbulence and the variability in particle fall speeds due to differences in shape and size.

2.3 MERRA2 reanalysis data

The Modern Era-Retrospective analysis for Research and Applications (MERRA-2), produced by NASA’s Global Modeling and Assimilation Office (GMAO), is a powerful reanalysis dataset for studying historical atmospheric data. Reanalysis involves the use of a consistent data assimilation system to reprocess meteorological observations over long periods, providing a coherent and reliable dataset, as described by Gelaro et al. (2017). By using an underlying forecast model, MERRA-2 integrates various sources of observations in a physically consistent way, even for variables that are sparsely or indirectly measured. The model operates with a horizontal resolution of approximately $0.5^\circ \times 0.65^\circ$ and incorporates 72 vertical levels, performing 3D-Var assimilation of data from a wide range of sources—such as surface stations, upper-air measurements, satellites, ships, and aircraft—at 6-hourly intervals (Kuma et al., 2021). For this study, we use the 3-hourly instantaneous assimilated meteorological fields (`inst3_3d_asm_Nv` (M2I3NVASM)) outlined

by Bosilovich et al. (2015) to generate simulated ceilometer profiles through the Automatic Lidar and Ceilometer Framework (ALCF). This dataset will serve as a critical point of comparison with our observational ceilometer data, enabling us to evaluate the effectiveness of the reanalysis in simulating cloud behavior in the troposphere.

2.4 Automatic Ceilometer and Lidar Framework (ALCF)

The ceilometer data collected by the Vaisala CL51 was pre-processed using the Automatic Ceilometer and Lidar Framework (ALCF), dedicated software developed by Kuma et al. (2021). This software was crucial for removing noise and resampling the raw data, which had been converted into NetCDF files. The noise removal process involves estimating the noise distribution at the highest available altitude range and subtracting its mean from all bins in the column, effectively filtering out unwanted signals (Guyot et al., 2022). Once the noise was removed, the data was resampled to ensure temporal consistency, making it suitable for further analysis.

Calibration of the data was another essential step. We implemented the calibration method and obtained the same value as the default value used by Kuma et al. (2021), which is 1.2×10^{-3} . This calibration method, based on O'Connor et al. (2004) lidar ratio approach, was specifically tailored for fully attenuating liquid stratocumulus clouds, ensuring accurate atmospheric measurements. Additionally, the ALCF software includes a cloud detection algorithm that employs an attenuated volume backscattering coefficient threshold to mask clouds in the data. For this study, we applied the default threshold of $2.6 \times 10^{-6} \text{ m}^{-1} \text{ sr}^{-1}$, which strikes a balance between minimizing false detections and avoiding missed observations in the Southern Hemisphere. This is particularly important due to the minimal influence of anthropogenic aerosols in this region (Kuma et al., 2021).

In addition to processing the ceilometer data, ALCF was also used to simulate MERRA-2 reanalysis data for comparison with the observed CL51 data. The simulation employed the lidar equation on model levels, calculating scattering and absorption by cloud particles and air molecules using Mie and Rayleigh theory, similar to the COSP lidar simulator method proposed by Chiriaco et al. (2006). However, ALCF incorporates improvements such as the ability to simulate Automatic Lidar and Ceilometer (ALC) with different operating wavelengths and a parameterization of backscattering from ice crystals based on temperature (Kuma et al., 2021). This comprehensive processing enabled us to generate daily NetCDF files that include the total attenuated volume backscattering coefficient at a resolution of 5 min and a bin vertical resolution of 50 m, which can be used to derive meaningful cloud statistics.

2.5 MRR2 data processing

In this section, we outline the processing steps for the Micro Rain Radar (MRR2) data, focusing on a specific range of gates between 400 meters and 3000 meters. The last gate is excluded from the analysis due to its high noise levels, as noted by Maahn and Kollias (2012). This initial filtering step ensures that the data we analyze reflects more accurate atmospheric conditions, enabling better representation of hydrometeors.

After selecting the appropriate range of gates, we apply the Hildebrand and Sekhon (1974) noise removal technique to enhance the quality of our Doppler spectra. This

statistical method systematically identifies and eliminates noise from the spectrum by sorting the spectral bins by amplitude. The algorithm iteratively removes the largest bins while calculating the mean and variance of the remaining spectrum. The process continues until the ratio of the squared mean to the variance exceeds a predetermined threshold. Once this condition is met, the identified bin serves as the noise limit, and the estimated noise level is subtracted from the observed spectral densities, isolating the backscatter signals generated by hydrometeors.

With the noise removed, we can calculate the effective reflectivity Z_e , which is crucial in radar meteorology as it represents the total reflectivity of hydrometeors. The calculation of the radar reflectivity factor η occurs at each height level using a combination of the raw spectral power matrix (**F_matrix**), the transfer function matrix (**T_matrix**), the calibration constant (**CC**), and the height matrix (**H_matrix**). The reflectivity factor is computed with the formula:

$$\eta = \frac{(\mathbf{F_matrix} \cdot \mathbf{CC} \cdot (i + 1)^2 \cdot \Delta H)}{(\mathbf{T_matrix}[1, i] \cdot 10^{20}\eta)} \quad (2)$$

where i represents the height index, and ΔH is the radar's vertical resolution. After obtaining η , the effective reflectivity Z_e is calculated by summing the reflectivity across all Doppler velocity bins:

$$Z_e = \frac{(10^{18} \cdot |K^2| \cdot \lambda^4 \cdot \sum \eta_3)}{(V_{\max} \cdot \pi^5)} \quad (3)$$

Here, $|K^2|$ is the dielectric constant (typically 0.92 for liquid water), λ is the radar wavelength, and $V_{\max}$ is the maximum Doppler velocity. The final reflectivity values are converted to logarithmic scale (dBZ) to produce the effective reflectivity matrix Z_e .

Following the calculation of Z_e , we derive the Doppler velocity, which indicates the speed of hydrometeors relative to the radar. This critical parameter is obtained from the cleaned Doppler spectrum after applying the noise removal technique. The cleaned spectrum reveals a peak that corresponds to the center of the hydrometeor distribution in the velocity domain. The Doppler velocity is calculated by locating this peak and extending the velocity range using the triple-spectra method. The formula for calculating the Doppler velocity is as follows:

$$V_{\text{matrix}}[i, j] = \text{triple_velocity}[\text{centre_index}] \quad (4)$$

where the **centre_index** corresponds to the strongest signal return in the cleaned spectrum. This process was illustrated in Fig. 1

Additionally, we compute the spectral width, which provides insight into the variability of hydrometeor motion and atmospheric turbulence. After removing noise, the algorithm identifies the points where the signal decreases on both sides of the central peak, with the spectral width determined by the distance between these points. This width in velocity units is calculated using the formula:

$$\text{Width_matrix}[i, j] = \text{width} \cdot V_\delta \quad (5)$$

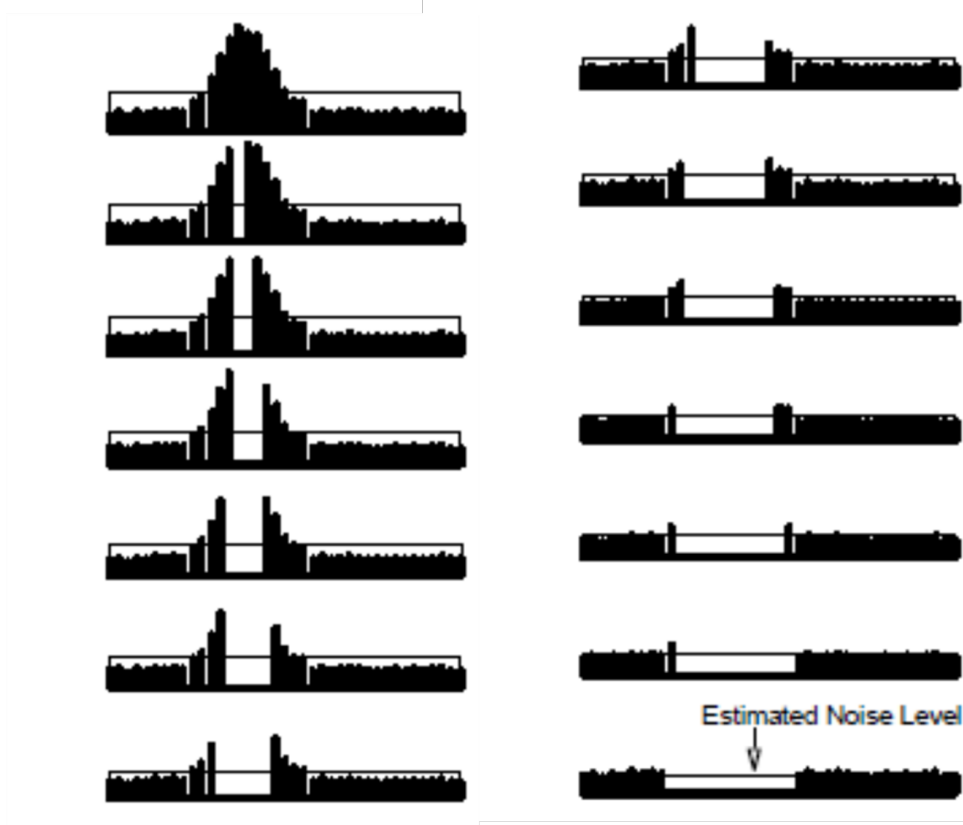


Figure 1: Schematic diagram illustrating the estimation procedure for noise levels. The thin horizontal lines represent the mean spectral power of the remaining spectral lines (METEK, 2010)

where V_δ is the radar’s velocity resolution. This spectral width provides important information about the distribution of particle velocities in the atmosphere.

From the effective reflectivity Z_e , we derive the snow rate (S_r) using the power law relationship which is:

$$S_r = \left(\frac{10^{(Z_e/10)}}{a} \right)^{\frac{1}{b}} \quad (6)$$

The coefficients a and b can vary depending on location and snow type. For this study, we utilize the $Z_e - S_r$ relationships established by Kulie and Bennartz (2009) and Matrosov (2007). These relationships were developed based on various snowfall particle shapes, allowing for accurate estimation of snowfall rates from radar data. Additionally, we consider the $Z_e - S_r$ relationship derived by Wiener et al. (2024), specifically tailored for the Dumont D’Urville station in East Antarctica. This approach allows us to examine the variations in relationships between different snow types and locations. All the relationships used in this study are summarized in Table 1.

Kulie and Bennartz (2009) derived several $Z_e - S_r$ relationships for an extensive range of snow particle habits and size distributions, providing global estimates of snowfall rates

Table 1: The Ze-Sr relationships used in these studies.

Ice Habit / Location	Work	Ze-Sr Relationship
Three-bullet rosettes	Kulie and Bennartz (2009)	$Ze = 24.04 \cdot Sr^{1.51}$
Aggregates	Kulie and Bennartz (2009)	$Ze = 313.29 \cdot Sr^{1.85}$
Low-density spheres	Kulie and Bennartz (2009)	$Ze = 19.66 \cdot Sr^{1.74}$
Aggregates spheroid	Matrosov (2007)	$Ze = 56 \cdot Sr^{1.2}$
Dumont D’urville stations	Wiener et. al (2023)	$Ze = 43.3 \cdot Sr^{0.88}$

from satellite radar data. They identified three relationships representing aggregates, three-bullet rosettes, and low-density spherical snow particles, which encapsulate the uncertainty range of their particle ensemble. All selected relationships apply to dry snowfall conditions, where snow particles remain largely unaltered by melting or riming processes, and their sizes are very similar to observed particles at various Antarctic locations. It is noteworthy that these relationships were established using a 35 GHz radar frequency, and the uncertainty associated with applying these $Ze - Sr$ relationships to a 24 GHz Micro Rain Radar (MRR) is minimal for small snow particles typically found in Antarctica, as indicated by Kneifel et al. (2011).

2.6 Synoptic Typing

Different synoptic weather patterns significantly influence cloud occurrence, leading to variations in precipitation and snow accumulation. This relationship is evident in the findings of Jolly et al. (2018), which illustrate how distinct synoptic regimes derived from synoptic-scale Coggins classifications, developed by Coggins et al. (2014) and Coggins and McDonald (2015), produce varying cloud occurrences. Specifically, the Strong Northern Cyclonic (SNC) regime exhibits the highest frequency of cloud formation, while the Weak Southern Cyclonic (WSC) regime shows the least. Understanding these patterns is crucial, as variations in cloud occurrence are likely to affect snow rates and overall snow accumulation, particularly in regions like Scott Base, Antarctica, where precipitation typically originates from these clouds.

In this study, we aim to explore the impact of synoptic-scale weather patterns on snow accumulation at Scott Base. To achieve this, we will adopt methodologies previously utilized in McDonald (2016) and McDonald et al. (2016). Our analysis employs the Self-Organizing Map (SOM) technique applied to the ERA5 dataset, spanning 30 years from 1993 to 2023. This approach allows us to derive representative surface wind patterns. Subsequently, we attribute outputs from other reanalysis datasets to these identified patterns using a Euclidean distance metric. The SOM is an iterative, unsupervised learning method widely used for clustering, as described by Kohonen (1990). During the learning process, a set of reference vectors is adjusted based on the differences between these vectors and the input data. A designated learning rate governs the extent of these adjustments, and the training continues for numerous iterations until the reference values stabilize. In each iteration, the reference vector that best matches the input data is identified and updated accordingly.

To optimize the analysis, we opted not to use the full 30 years of ERA5 output directly with the Self-Organizing Map (SOM). Instead, we adopted a method used in McDonald

et al. (2024) by initially conducting Empirical Orthogonal Function (EOF) analysis on the space-time cube representing surface wind data. This preliminary step allowed us to focus the SOM application on the most significant principal component (PC), thereby improving the model's efficiency. By truncating the set of PCs when the explained variance reached 90% of the total variance, we streamlined the data processing. The SOM methodology was implemented using the mini-SOM Python package for this study. Subsequently, we correlated the outcomes of our trained SOM with daily wind profiles for the year we are investigating. This correlation facilitated the classification of weather patterns associated with each specific day in our analysis.

Following the training of our SOM, we will correlate the results with daily wind profiles from the selected year under analysis. This step will enable us to classify the prevailing weather patterns for each day, offering insights into how specific synoptic weather conditions relate to snow accumulation patterns at Scott Base. By understanding these connections, we can better predict and analyze snowfall in this critical region, contributing to our broader understanding of climate dynamics in Antarctica.

3 Result

In this section, we present the results obtained by applying the analysis technique described in Section 2 to our CL51 ceilometer data and MRR2 radar data collected at Scott Base, Antarctica. The analysis aims to provide insights into the observed patterns. We begin with Sections 3.1 until 3.4, which focus on the lidar and radar measurements profiles and climatology, respectively. This is followed by Section 3.5, where we compare the lidar and radar climatology side by side to examine how different types of cloud content, based on the strength of the attenuated volume backscattering coefficient, can produce varying types of snowfall. Finally, Section 3.6 wraps up the results by classifying snow accumulation based on different weather states.

3.1 Attenuated volume backscattering profile

In this section, we present the results obtained after preprocessing and analyzing the Vaisala CL51 ceilometer data using ALCF software. Figure 2 illustrates a time-altitude plot of the attenuated volume backscattering coefficient ($\text{m}^{-1}\text{sr}^{-1}$) for two consecutive days, May 5th (a) and 6th (b), 2022. Throughout the observation period, the cloud base heights remain generally low, with most cloud structures masked out at around 2 km altitude.

On May 5th, a thin strip of elevated attenuated volume backscattering coefficient is observed between 00:00 UTC and 18:00 UTC at approximately 2 km altitude. This layer, depicted in a yellowish hue, indicates regions with stronger backscattering compared to the surrounding areas. Some portions of this layer are masked as clouds between 00:00 UTC and 02:00 UTC, suggesting denser cloud cover in those regions. Below this high backscattering cloud layer, a broader area of green-coloured backscattering signals extends from 03:00 UTC to around 19:00 UTC, which is likely indicative of precipitation occurring beneath the cloud base. Notably, fluctuations in the backscattering coefficient can be seen below the yellow strip, implying variability in atmospheric conditions at lower altitudes.

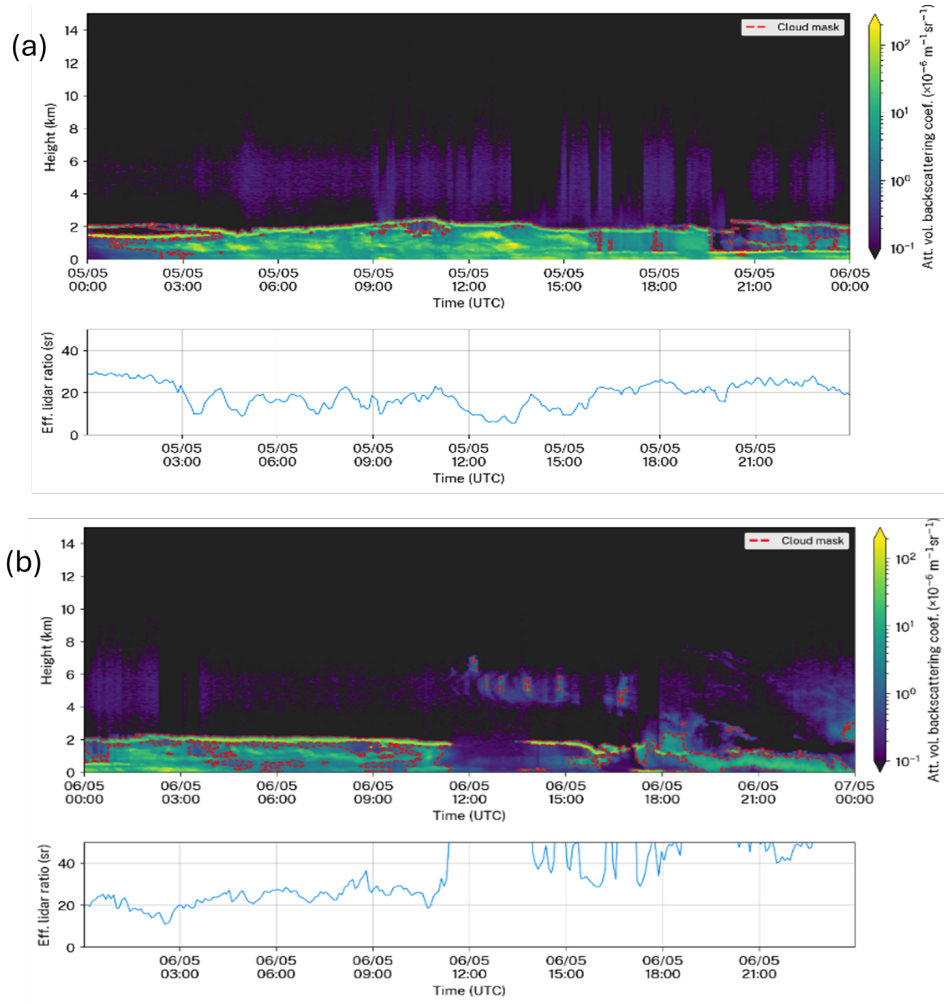


Figure 2: The time-altitude plot of the attenuated volume backscattering coefficient is analyzed from the Vaisala CL51 ceilometer data collected on May 5, 2022 (a), and May 6, 2022 (b).

Additionally, isolated patches with higher backscattering coefficients appear within this lower region. As the day progresses, a thin layer of high backscattering signal is detected near the surface between 21:00 UTC and 23:59 UTC, indicating the presence of near-surface atmospheric phenomena.

The observations on May 6th, 2022, show similar patterns, with another thin layer of highly attenuated backscattering signals observed between 00:00 UTC and 11:00 UTC at around 2 km altitude. Interestingly, low backscattering coefficients are also masked as clouds at this altitude, suggesting that the cloud composition may differ from that of the thin layer with high backscattering values.

Throughout both days, the lidar ratio exhibits notable variations: it starts high at around 20 sr, decreases to approximately 10 sr by 03:00 UTC, and then fluctuates between 10 and 20 sr before rising back to around 20 sr by 23:59 UTC. These observations highlight the complex atmospheric dynamics captured by the ceilometer over the two-day period.

Using the methodology outlined in Section 2.6, we successfully simulated pseudo data from the MERRA2 reanalysis product to compare with the ceilometer observations. Fig-

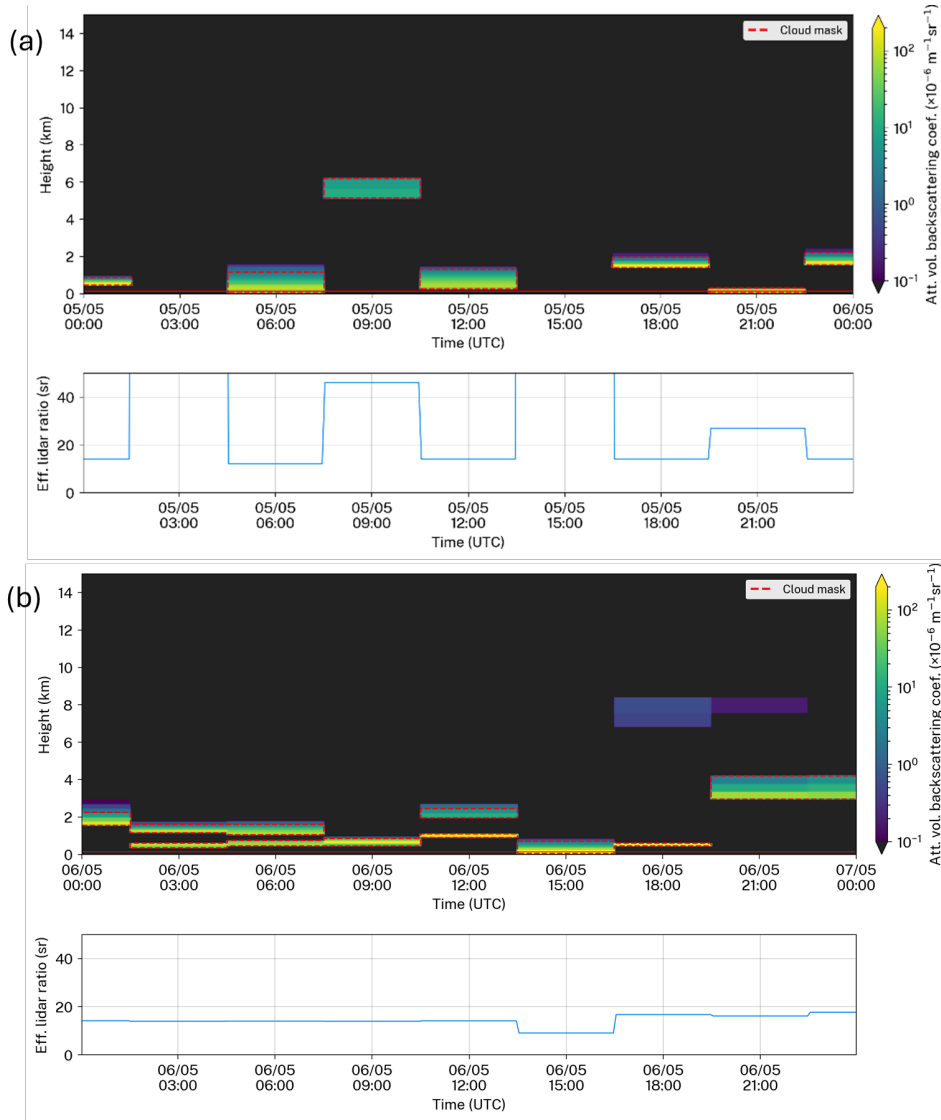


Figure 3: The time-altitude plot of the attenuated volume backscattering coefficient is analyzed from the MERRA2 reanalysis data collected on May 5, 2022 (a), and May 6, 2022 (b).

Figure 3 displays a time-altitude plot of the attenuated volume backscattering coefficient derived from the MERRA2 reanalysis for May 5th and 6th, corresponding to the observations shown in Figure 2.

On May 5th, we identified that the majority of the cloud layers were concentrated below 2 km altitude, exhibiting varying cloud thicknesses and backscatter signals. Within these clouds, the backscatter was strongest at the cloud base and gradually diminished as it approached the cloud top, especially evident between 05:00 and 07:00 UTC when thicker clouds, ranging from approximately 500 to 1000 meters in thickness, were present. The backscatter coefficient started around $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$ at the base and decreased to around $10^{-5} \text{ m}^{-1} \text{ sr}^{-1}$ at the top. Additionally, we detected a high-altitude cloud layer around 5 km with a weaker backscattering coefficient of about $10^{-5} \text{ m}^{-1} \text{ sr}^{-1}$.

A comparison between the observations and the model plot from Figures 2(a) and 3(a)

revealed that the model attempted to simulate the low-level clouds observed but underestimated their density and extent. While the observations showed a persistent cloud layer at approximately 2 km altitude, the model indicated alternating periods of cloudy and clear skies.

Turning to the results for May 6th, the reanalysis data showed a high backscatter signal associated with a cloud that gradually increased in altitude from near ground level to about 1 km between 02:00 and 13:00 UTC. This cloud maintained a relatively stable backscattering coefficient of around $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$ throughout its ascent. Additionally, a secondary cloud layer appeared above a thinner yellow cloud layer, spanning from 00:00 to 07:00 UTC.

When comparing the observational data with the reanalysis results for May 6th, it became evident that the model was able to simulate some of the observed cloud dynamics between 00:00 and 11:00 UTC but struggled to accurately represent the thin yellow cloud layer seen in Figure 2(b). Furthermore, the model predicted a thick cloud layer at around 3 km with a backscattering coefficient ranging from $10^{-4.5} \text{ m}^{-1} \text{ sr}^{-1}$ that gradually decreased, which did not appear in the observational data. This discrepancy highlights the model's difficulty in accurately capturing the tropospheric columns at Antarctica.

3.2 Lidar climatology

Shifting focus from short-term fluctuations to long-term trends, we transitioned from analysing daily data to examining yearly patterns, allowing for a broader discussion of climate behaviour. Based on the analysis of seasonal cloud patterns throughout 2022, the cloud occurrence was examined by comparing observations from the Vaisala CL51 ceilometer and MERRA2 reanalysis data. Figure 4 illustrates the variations in cloud occurrences across different seasons and altitudes. Overall, both the observational and reanalysis data indicate that low-altitude clouds, particularly around 2 km, have the highest percentage of occurrence, with a gradual decrease in cloud presence as the altitude increases.

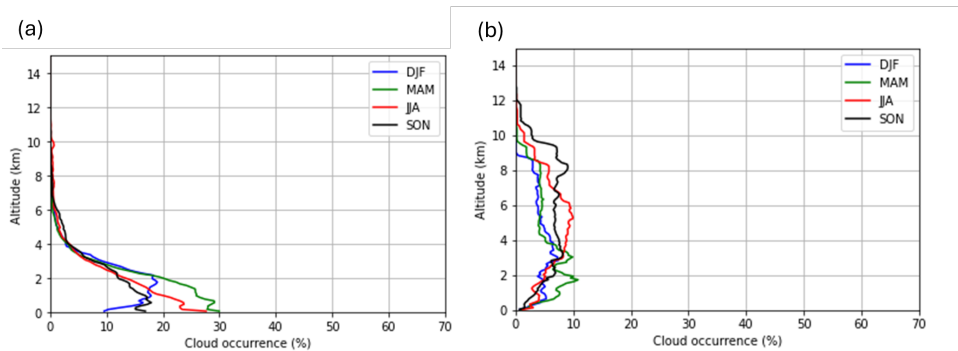


Figure 4: The cloud occurrence in 2022 based on seasonal variations, with panel (a) showing observational data from 2022 and panel (b) displaying the MERRA2 model data for the same year.

However, the MERRA2 model data shows a slower increase in cloud occurrence, starting from no occurrence at the ground level and gradually rising to a peak of around 10% at 2 km altitude. From there, the model data shows a steady occurrence of clouds up

to about 8 km, after which the cloud presence gradually decreases to 0% as altitude increases. This contrasts with the observational data, where the decline in cloud occurrence is more gradual, dropping from around 30% to 0% over a 4 km interval.

Focusing on the observations from the Vaisala CL51 ceilometer data, we found that seasonal variations in cloud occurrence differ within a 20% range, with most of these variations occurring near ground level, within 1 km of altitude. Beyond this, as we reach 3 km in altitude, the seasonal differences in cloud occurrence begin to smooth out. A significant amount of ground-level cloud detection was observed, which we suspect may be due to fog or blowing snow. During the austral autumn (MAM), the highest cloud occurrence below 2 km altitude was recorded, reaching around 20%. The austral summer followed closely, with the second-highest occurrence of approximately 19% at 2 km altitude, maintaining this value as altitude decreases. The lowest cloud occurrence at this level was recorded during austral spring and winter, reaching approximately 10%.

When examining the MERRA2 reanalysis data, there are no obvious variations between the seasonal plots. The model shows lower cloud occurrences below 2 km, with most of the cloud presence in this region remaining below 10%. In comparison, the observations clearly indicate that the model significantly underestimates the amount of cloud cover at mid- to low-levels of the troposphere. Notably, in terms of seasonal variations, the highest observable cloud altitudes during summer reach around 9 km, while in winter and spring, cloud occurrences extend slightly higher, up to approximately 12 km. At higher altitudes within the tropospheric column, the model data tends to overestimate cloud amounts. While the model suggests clouds can occur up to 12 km, the highest clouds observed in the actual measurements are generally around 9 to 10 km with extremely low occurrence.

3.3 Radar measurement profile

In this section, we applied the method outlined in Section 2.5 to analyse the MRR2 radar data, focusing on snowfall events at Scott Base, Antarctica. Figure 4 presents various aspects of this analysis for data collected on April 4, 2022, including the effective reflectivity (Z_e), Doppler velocity, spectral width, and the derived snow rate. These parameters help to understand the snowfall patterns and dynamics throughout the day.

Figure 5(a) shows the effective reflectivity, with a strong signal detected between 00:00 UTC and 12:00 UTC. The highest reflectivity forms a strip-like shape along the 1 km altitude, with values ranging from 15 to 20 dBZ. While other parts of the plot also show reflectivity signals, these are less consistent. Some signals extend down to ground level, particularly around 06:00 UTC and from 10:00 UTC to 12:00 UTC. The strip-like structure of the reflectivity suggests that we are observing a vertically moving phenomenon, indicative of snowfall occurring during this period.

The snowfall observation is further supported by the Doppler velocity data shown in Figure 5(b). The Doppler velocity provides insight into the movement and speed of the hydrometeors, which in this case is snow. Positive Doppler velocities indicate objects moving towards the detector, while negative velocities represent objects moving away. For most of the observed period, the snow appears to be falling towards the detector. However, between 03:00 UTC and 05:00 UTC, there is a noticeable updraft, where the snow is momentarily carried upwards. Additionally, the region along the 1 km altitude shows a lighter red intensity, suggesting that precipitation is descending more slowly in

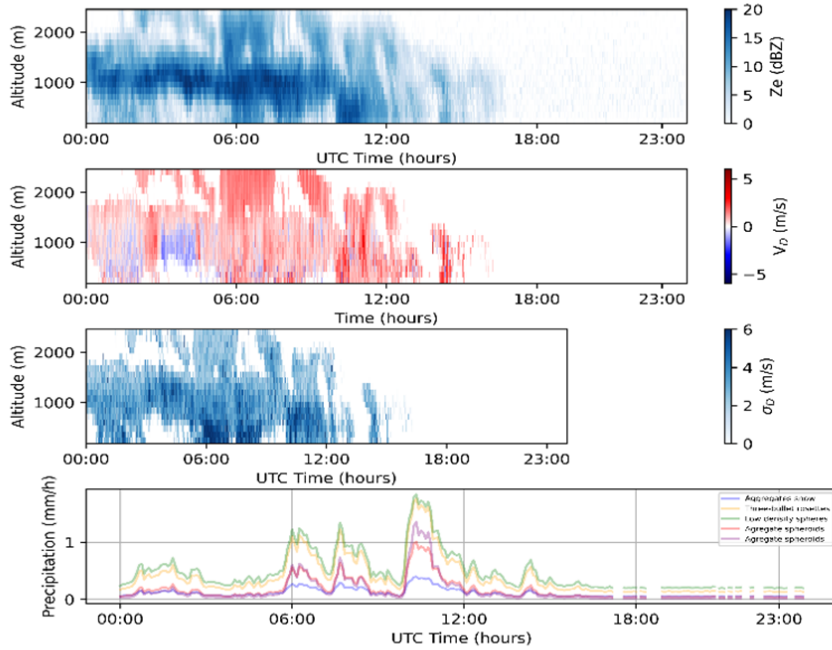


Figure 5: Radar climatology results from observations on April 4, 2022, showing (a) effective reflectivity (Z_e), (b) Doppler velocity, (c) spectral width, and (d) precipitation rate.

this area. This coincides with a period of reduced precipitation reaching the surface, likely due to the influence of the updraft.

Next, we examine the spectral width plot in Figure 5(c), which provides information about the turbulence and variability within the falling snow particles. The spectral width reveals broader values between 00:00 UTC and 12:00 UTC, especially around the 1 km altitude region. This indicates more turbulence or variability in the falling snow particles during this period, which could be related to varying wind conditions or the presence of snowflakes with different sizes and shapes. The increased spectral width aligns with the observations of the updrafts and suggests a dynamic environment where air movements are affecting the falling snow.

Lastly, we analysed the snow rate or precipitation rate on April 4, 2022, as shown in the bottom panel of Figure 5. The highest precipitation spikes occurred around 06:00 UTC, 08:00 UTC, and the peak around 11:00 UTC, with rates reaching up to 1 mm/h, depending on the Z_e - S_r relationship used, as detailed in Table 1. These spikes correspond to periods of stronger reflectivity and increased snowfall intensity, illustrating the correlation between radar measurements and surface precipitation rates. The analysis provides a comprehensive view of the snowfall event, from its vertical structure to the resulting precipitation on the ground.

3.4 Radar climatology

This analysis explores the variations in reflectivity and Doppler velocity data across different altitude levels, focusing on how these measurements change from the surface to 3000 meters. Figure 6 presents the median values (solid black line) alongside the 25th and 75th percentiles, forming the interquartile range (IQR, dashed blue lines) for reflectivity and Doppler velocity. The data used in this analysis was carefully reprocessed, considering only periods where at least 15 of the 23 available data points were valid, with invalid points (represented as NaNs) excluded. This approach ensures a robust dataset for assessing the vertical structure of the atmosphere, as snow precipitation events, such as storms, were infrequent during the analysis period.

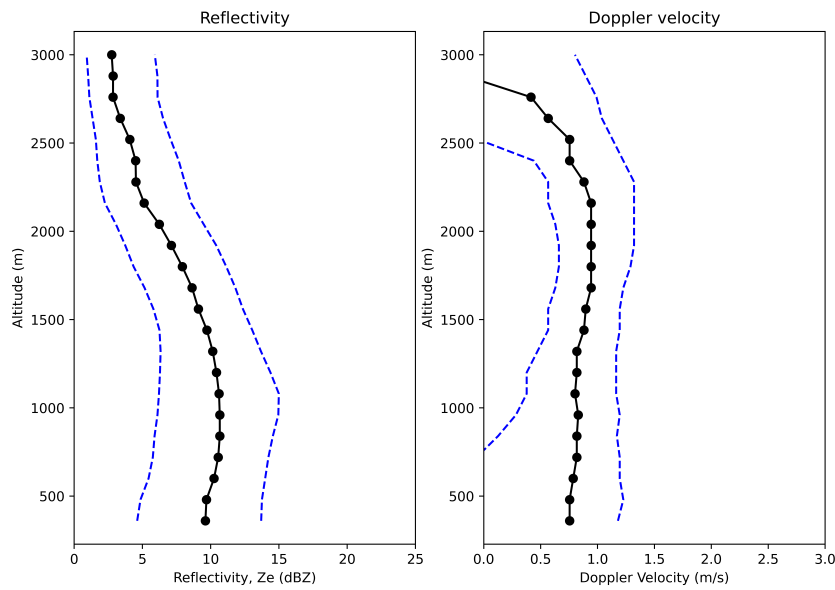


Figure 6: The median (solid black line) and the 25th and 75th percentile interquartile range (dashed blue lines) for reflectivity and Doppler velocity.

The median reflectivity profile shows a smooth decline with increasing altitude, reflecting the diminishing concentration of precipitation particles such as snowflakes and ice crystals. Near the surface, median reflectivity values range from 10 to 15 dBZ, indicating a relatively denser presence of snow. However, this value gradually drops to about 5 dBZ at around 3000 meters, highlighting a reduction in particle concentration at higher altitudes. The IQR for reflectivity is wider near the surface, suggesting greater variability in precipitation intensity or particle composition. As altitude increases, the IQR narrows, pointing to a more uniform distribution of snow particles at these levels.

The Doppler velocity profile reveals the motion of snow particles toward the radar, with positive values indicating movement toward the observer. The median Doppler velocity is approximately 1 m/s near the surface, suggesting a moderate speed of particle movement. This value gradually decreases to below 0.5 m/s at higher altitudes, likely reflecting changes in wind speed or the distribution of falling snow particles. The IQR for Doppler velocity is narrow near the surface, indicating less variability in particle speeds,

but becomes wider with increasing altitude. This broadening suggests a more diverse range of particle velocities at higher levels, potentially due to varying wind patterns or increased atmospheric turbulence. The smooth median line and expanding IQR at altitude illustrate the complex interactions between wind and precipitation in the vertical profile

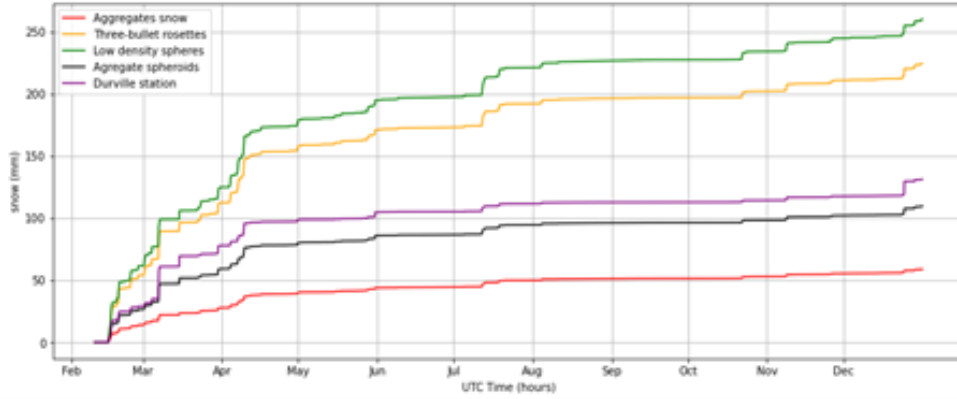


Figure 7: Snow accumulation estimates from 2022 radar observations, calculated using five different Ze-Sr relationships.

We conducted an analysis to derive snow accumulation from the snow rate values, providing valuable insights into the variations in snowfall throughout the year. The methodology for these derivations is detailed in Section 2.4. In Figure 7, we present daily snow accumulation variations throughout 2022 at Scott Base, Antarctica, utilizing different Ze-Sr relationships to illustrate these patterns. The details of this relationship can be found in Table 1.

The analysis shows that low-density spheroids (green line) recorded the highest snow accumulation over the entire year, reaching a total of 250 mm. Three-bullet rosettes followed as the second highest, with a total accumulation of around 180 mm. The D’Urville Station data and aggregate spheroids exhibit nearly similar accumulation patterns, with the D’Urville Station data recording approximately 20 mm more snow than the aggregate spheroids. In contrast, aggregate snow had the lowest accumulation throughout the year, reaching only about 50 mm.

The most significant snow events occurred from summer to autumn (February to May), where steep increases in snow accumulation were observed, particularly for low-density spheroids and three-bullet rosettes, which reached up to 150 mm, while other snow types remained below 100 mm. After May, as winter and spring approached, the snow accumulation slowed, showing more stagnant but gradual increases. A slight rise in snowfall is observed again in October, though the accumulation rates remain lower compared to the earlier part of the year. The data from the D’Urville Station (purple line) displays less variability and a steadier accumulation trend compared to the model-derived data for other snow types.

3.5 Lidar and radar profile comparison

In this section, we compare ceilometer lidar and radar observations to explore the relationship between clouds and snowfall. We employed a straightforward comparison method

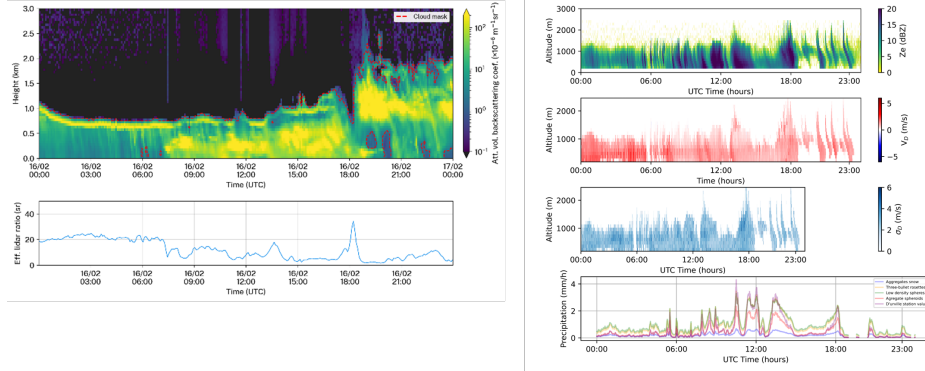


Figure 8: Comparison plot between lidar and radar derived profile for 16 February 2022.

by analyzing the climatological data from both instruments, with the initial goal of understanding how likely supercooled liquid water content (SLWC) is to produce heavy snowfall. We focus here on key features that offer insight into the relationship between clouds and snow, and assess how effectively lidar and radar observe the vertical structure of the Antarctic troposphere.

Figure 8 illustrates the climatological data from lidar and radar taken on February 16, 2022, with both datasets constrained to an altitude range of 0 to 3000 meters to maintain consistency in scale. Generally, both instruments recorded a high backscattering coefficient and reflectivity values between 1 to 1.5 km and down to ground level throughout the day, suggesting a persistent snowfall event, potentially even a snowstorm. Between 00:00 UTC and 07:00 UTC, a thin layer of highly attenuated backscattering coefficients around $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$ was observed, hinting at the presence of SLWC. A more detailed analysis of this phenomenon is discussed in Section 4. Beneath this layer, a relatively low and consistent backscattering was noted, indicating possible precipitation. Radar data for the same period revealed moderate reflectivity values ranging from 5 to 10 dBz, aligning with the lidar observations and confirming a snowfall event below 1 km during this time frame.

One interesting observation is that the radar data did not capture the thin, highly backscattered cloud layer at 1 km altitude seen by the lidar. Additionally, both instruments recorded a substantial snowfall event from 09:00 UTC to 18:00 UTC, evidenced by a high backscattering coefficient in the lidar data and corresponding high radar reflectivity ranging from 15 to 20 dBz. This period coincided with a peak in precipitation rates, reaching up to 4 mm/h. Notably, while the lidar data showed minimal variation in the backscattering coefficients during this time, the radar data displayed variations with alternating regions of high (around 20 dBz) and lower reflectivity, indicating different snowfall intensities. Furthermore, the radar revealed vertical stripes of high reflectivity around 18:00 UTC to 23:59 UTC, suggesting thin snowfall bands. This observation was mirrored in the lidar data, which recorded dense cloud formations, as indicated by high backscattering coefficients.

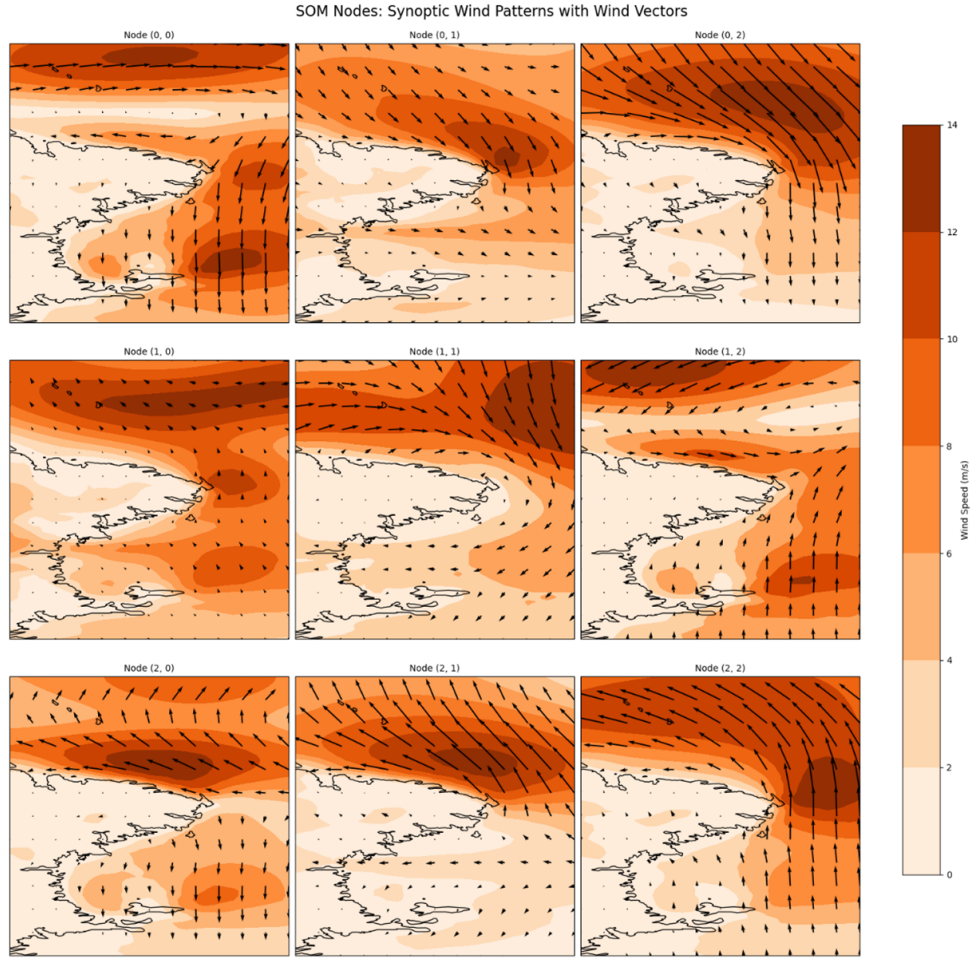


Figure 9: A 3×3 Self-organising map weather states with each node representing different weather state derived from method in Section 2.5.

3.6 Weather state classification

Using the method discussed in Section 2.5, we classified weather patterns derived from surface wind data using a Self-Organizing Map (SOM). This approach allowed us to identify various atmospheric conditions that contribute to snow accumulation patterns around Scott Base, Antarctica. Figure 9 illustrates the different weather states, with each node representing a unique pattern. The region around Scott Base is influenced by both polynya and open sea conditions, which impact precipitation and snowfall patterns. This classification provides insight on how changes in surface wind dynamics affect snowfall and other meteorological phenomena in the area. The results for snow accumulation classifications are shown in Figure 10.

From Figure 10, it is evident that snow occurrence is notable only in nodes with more than 10% occurrence: node (0,0) shows 14.24%, node (0,1) has the highest occurrence at 43.05%, and node (1,1) follows with 31.13%. The weather pattern represented by node (0,0), as seen in Figure 9, is characterized by anomalous flow, which leads to a low frequency of occurrence. Node (0,1) is associated with a low-pressure system to the north of the domain, where wind flows from areas of high pressure toward the low-pressure zone. This transition is accompanied by a decrease in wind speed as it moves northward. Addi-

tionally, we observed strong katabatic winds descending from the Transantarctic Mountain region, flowing from west to east. The weather pattern in node (1,1) is characterized by a low-pressure system to the northeast, where the wind bends westward, directing flow toward the Antarctic landmass. In the vicinity of Scott Base, wind speeds are relatively low, ranging from 0 to 4 m/s.

In Figure 10, we observed nodes with almost no occurrence, specifically node (1,0), node (1,2), and node (2,0). Node (1,0) exhibits quiescent conditions characterized by low synoptic forcing, where most wind components display weak magnitudes and lack distinct directions. This leads to a relatively calm atmosphere, suggesting that there is less moisture transported into the atmosphere to produce snowfall. Node (1,2), on the other hand, shows the interaction between the Ross Air Stream (RAS) and anticyclonic circulations, which subsequently moves toward a low-pressure region in the east. The RAS is a dominant northward stream of air over the Ross Sea, facilitating significant transport of atmospheric mass from the interior of Antarctica (Seefeldt and Cassano, 2012). Meanwhile, node (2,0) is associated with a low-pressure system to the northeast of the domain, featuring a northerly flow at a moderate speed.

Lastly, we observed two nodes with snow occurrence ranging from 1% to 5%: node (2,1) with 2.98% occurrence and node (2,2) with 1.66% occurrence. Both nodes exhibit similar weather patterns characterized by a very strong Ross Air Stream (RAS); however, they differ in the magnitude and slight variations in the direction of the wind in the middle of the domain. Specifically, node (2,2) has a higher wind speed magnitude compared to node (2,1), with moderately flowing winds directed toward the Antarctic land. This difference in wind characteristics may explain why node (2,1) has a slightly higher occurrence of snowfall than node (2,2).

Focusing on the snow accumulation patterns in the highest occurrence node, node (0,1), we observed distinct variations in snow types, alongside one station location from different Ze-Sr, as noted in Table 1. In this node, low-density spheres—characterized by their unique formation and reduced density—accumulate a total of 500 mm of snow. This is followed by three-bullet rosettes, which accumulate nearly 400 mm. The other snow types display similar patterns, with accumulation values around 100 mm. Node (1,1) has the second-highest occurrence at 31.13%, with the highest cumulative snow recorded at 300 mm. Across all nodes with significant snow accumulation, low-density spheres and three-bullet rosettes dominate, while the remaining relationships exhibit nearly identical levels of accumulation. The green line, representing aggregates, consistently indicates the highest snow accumulation in nodes with substantial snowfall. Additionally, nodes with higher occurrences generally show greater variability in snow accumulation over time.

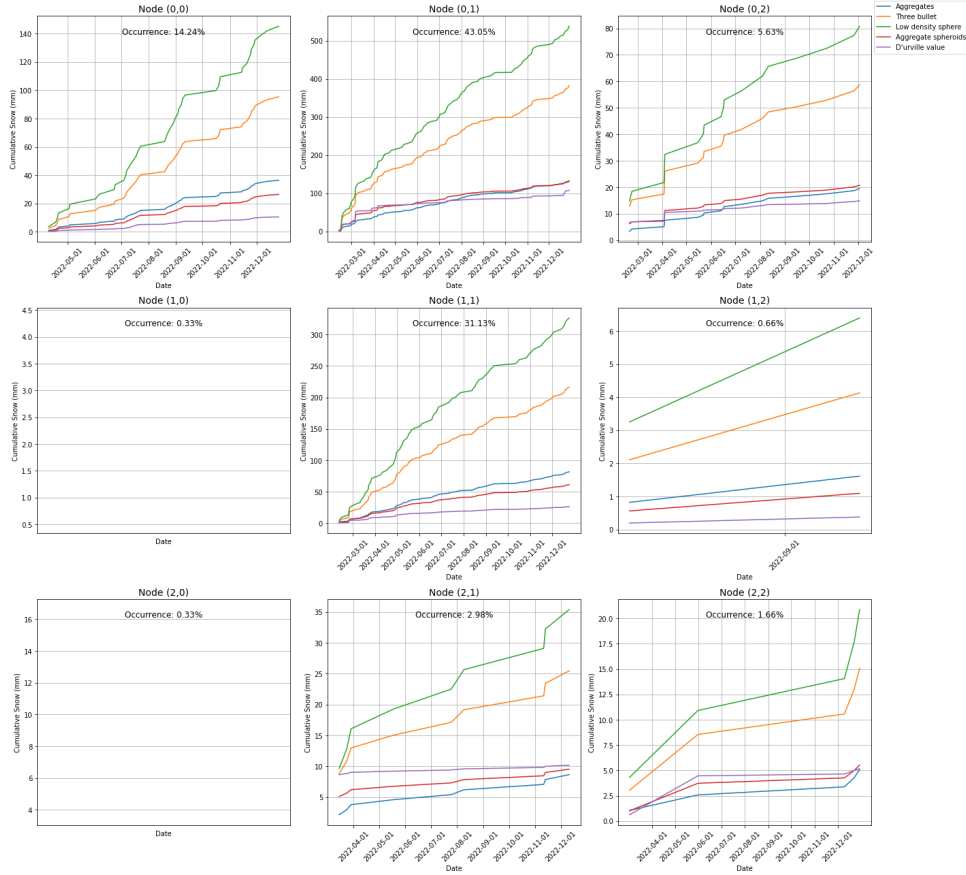


Figure 10: Snow accumulation patterns in 2022, classified according to synoptic weather patterns.

4 Discussion

In this study, we utilized the Automatic Ceilometer and Lidar Framework (ALCF) software, following the methods proposed by Maahn and Kollias (2012), to derive measurements from the Vaisala CL51 ceilometer and MRR2 radar. Both instruments have unique strengths in detecting atmospheric particles, offering different perspectives on hydrometeor measurements. The lidar is particularly effective at detecting small, concentrated particles like water droplets and ice nuclei in the atmosphere, as highlighted by Aubry et al. (2024). This capability is clearly demonstrated in the lidar backscatter profile shown in Figure 8, where the structure of the clouds is well-preserved. However, lidar struggles to accurately capture snow precipitation events due to the irregular structure of snowflakes, which scatter light unevenly, leading to potential biases. This bias is evident in Figure 8, where high backscatter values are observed beneath the cloud layer, typically indicating the presence of cloud particles or hydrometeors. However, the intensity of snowfall cannot be directly inferred from these backscatter measurements alone, as backscatter strength may be influenced by factors such as particle size, shape, and concentration. This introduces a bias in understanding precipitation formation, where high backscatter may not necessarily correspond to heavy snowfall. In contrast, radar instruments excel at measuring snow particles due to their sensitivity to particle size, making them more effective at detecting ice particles than water droplets, as further supported by Aubry et al. (2024).

This is evident in Figure 8, where the radar measurement profile shows a more detailed precipitation structure compared to the lidar profile.

In Section 3.1, we focused on the lidar-derived attenuated volume backscattering coefficient profile, analyzing cloud formations observed in Figure 2, particularly the presence of thin, highly backscattering cloud layers. These clouds, characterized by dense concentrations of water droplets, produce strong backscatter signals, suggesting the presence of supercooled liquid water (SLW). This observation aligns with studies such as Standford et al. (2023), who detected similar clouds at Macquarie Island, recording backscatter values exceeding $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$. Our findings are consistent with theirs, as well as with those of Guyot et al. (2022) at Davis, East Antarctica, and Whitehead et al. (2024) at Christchurch, New Zealand, both of whom reported comparable backscatter signals for SLW clouds. These similarities further confirm that the cloud formations discussed earlier were indeed supercooled liquid water clouds (SLWC).

In our comprehensive analysis of the entire year of 2022, we observed profiles of mid- and high-altitude clouds that exhibited notable characteristics. These clouds displayed a greater thickness alongside a lower backscatter value, measuring approximately $10^{-5} \text{ m}^{-1} \text{ sr}^{-1}$ or lower. This description aligns well with the findings of Gorodetskaya et al. (2015), who reported similar ice layers in the atmosphere, with backscatter values ranging from between $10^{-6.5} \text{ m}^{-1} \text{ sr}^{-1}$ and $10^{-5} \text{ m}^{-1} \text{ sr}^{-1}$. Furthermore, recent studies by Guyot et al. (2022) and Whitehead et al. (2024) corroborate these findings, as their ice cloud classification results also identify clouds within this backscatter range. This consistency across multiple studies underscores that our findings indeed identify these clouds as ice clouds.

We then shifted our analysis to examine the lidar climatology, revealing seasonal variations in cloud occurrence, as shown in Figure 4(a). Our findings indicate that cloud occurrence is most frequent during the autumn and summer, and least common in the spring and winter at 2 km altitude, suggesting that seasonal temperature changes play a significant role in cloud formation, particularly in coastal areas like Scott Base. The rise in surface temperatures during the summer likely warms the Southern Ocean, increasing atmospheric moisture through evaporation, which is then transported via advection, leading to cloud formation in the Antarctic region. These observations align with the results of Adhikari et al. (2012), which showed the highest cloud occurrence during the summer, with nearly 80% cloud cover around sea regions and approximately 50% near Scott Base, Antarctica. Our results are also consistent with Jolly et al. (2018), who observed a decrease in cloud occurrence with increasing altitude, noting a peak during the summer and autumn based on data from the Ross Ice Shelf (RIS) and Ross Sea (RS). However, unlike our findings, their study reported a higher frequency of high-altitude clouds during winter. This discrepancy could be due to differences in observational methods, as Jolly et al. (2018) used satellite-based data, while our study relied on ground-based remote sensing techniques.

In the results section, we analyzed and compared observational data with the MERRA2 reanalysis model, focusing on cloud occurrence as depicted in Figure 4. Our findings indicated that MERRA2 significantly underestimates cloud presence at lower altitudes (below 3 km) while overestimating it at higher altitudes (above 5 km). This pattern aligns with the study by McDonald et al. (2024), which also identified similar discrepancies between observed data and MERRA2 outputs. These biases may stem from the model's limited ca-

capacity to accurately capture the unique dynamics of the Antarctic tropospheric column, especially at lower levels. Previous research, such as that by Kuma et al. (2021), also highlighted MERRA2’s reduced accuracy in cloud representation, showing considerable variation when compared to ground-based observations.

In Section 3.4, we delve into the radar-derived climatology of snowfall, providing insights into the distribution and behaviour of snowfall events over time. Our analysis revealed that the median radar reflectivity tends to exhibit stronger signal intensities in the lower part of the vertical column, particularly around 1,000 meters, with a gradual decrease in reflectivity as altitude increases, as shown in Figure 6. This pattern indicates that snowfall is more concentrated at altitudes below 1.5 km. Our findings align with the results from Wiener et al. (2024) at Dumont D’Urville, East Antarctica, where their study also demonstrated a decline in median reflectivity with altitude, recording the highest values below 1 km. Additionally, Alexander et al. (2023) supports this trend, highlighting that the highest cumulative snowfall was observed around 1 km altitude. These similarities could be attributed to the coastal geographical positioning of both study areas, which influences snowfall characteristics.

The analysis also explored the variability of reflectivity with altitude through the interquartile range, which becomes narrower as altitude increases. This trend is likely due to the decreasing sensitivity of the Micro Rain Radar (MRR) with height, as described by Maahn and Kollias (2012). Moreover, Figure 7 illustrates the variations in snow types and the Ze-Sr (reflectivity-snowfall rate) relationships across different locations. Among the observed snow types, low-density spheres and three-bullet rosettes dominated the snow accumulation, consistent with previous discussions. In Antarctica, snow particles are generally smaller than those in other regions globally, as noted by Souverijns et al. (2017). Larger snow particles, with diameters reaching up to 10 mm, are more commonly found near coastal areas where increased water vapor availability influences their formation (Konishi et al., 1990). This emphasizes that location plays a significant role in determining snowfall types and rates, as varying geographic features impact cloud formation and precipitation processes.

Throughout the year, the analysis revealed that snowfall accumulation peaks during the summer, coinciding with the season’s steepest accumulation rates, and then gradually decreases as the year progresses. This seasonal trend can be attributed to moist air being advected into the atmosphere as the surrounding sea warms during summer. When comparing our total annual accumulation data to that of Cohen and Dean (2013) for the Ross Ice Shelf (RIS) region, particularly at Mary and Linda stations near Scott Base, Linda Station recorded around 200 mm of snow in 2012, while Mary Station saw approximately 500 mm in 2008. Although the result in this study might not perfectly represent our findings as they take into account factors like ablation, compaction, and sublimation causing negative accumulation, it provides valuable context in evaluating the accuracy of our results in capturing snow variations.

In Section 3.5, we compared the climatological data from both lidar and radar on specific days and identified a persistent snowfall event throughout the entire period, as shown in Figure 7. Lidar observations revealed that clouds composed of water droplets produce notably high backscattering values. This suggests that the thin cloud layers we detected were likely supercooled liquid water clouds (SLWC) with precipitation occurring beneath them. Previous research by Siliber et al. (2018) supports this interpretation, as

they found that SLWC tends to be confined within the lower 4 km of the atmospheric column. Thus, our observations align with these findings, indicating the presence of SLWC on those specific days. From the radar data presented in Figure 8, we further identified the continuous precipitation throughout the day, which aligns with the process of sustained nucleation and ice crystal growth, as discussed by Silber et al. (2018).

The intensity of the observed snowfall was notable, with precipitation rates surpassing the threshold of 1 mm/h, which is often used to distinguish between light and moderate precipitation (Rasmussen et al., 1999). In our case, the precipitation rate reached as high as 4 mm/h, confirming that this was indeed a significant snowstorm event. Additionally, the total accumulation observed that day exceeded the threshold for large accumulation events specified by Gorodetskaya et al. (2015), which is 10 mm, with our observations recording a substantial 20 mm. These findings indicate that heavy snowfall events can occur in coastal Antarctica, leading to significant accumulation during extended periods. The implications of these findings contribute to the understanding of snow dynamics in the region and enhance our knowledge of how moisture sources influence snowfall processes in Antarctica.

In Section 3.6, we presented cumulative snow classified by weather, revealing that node (0,1) dominates with a synoptic forcing occurrence of 43.01%, accounting for nearly half of the precipitation events for that year. As previously discussed, node (0,1) is associated with a low-pressure system to the north of the domain. An important factor in the frequency of precipitation events is the transport of moisture in the atmosphere. Periods with abundant cloud cover, particularly liquid clouds, indicate significant moisture transport, which may correlate with increased precipitation events. A study by Silber et al. (2018) showed that 75% of detected supercooled liquid water cloud (SLWC) layers over McMurdo Station precipitate from the liquid cloud base in the form of ice. This occurs due to the riming process, where ice-nucleating particles (INPs), such as snowflakes, combine with SLWC and grow into snow. This process facilitates the efficient growth of supercooled liquid water (SLW) into snow, rather than allowing snow to clump together to produce more snowfall. Therefore, it is essential to examine the synoptic forcing that correlates with increased cloud occurrence below 4 km altitude and heightened riming frequency. Cloud occurrence classified by synoptic regimes in Jolly et al. (2018) indicated that the Weak Northern Cyclonic (SNC) regime, characterized by relatively weak cyclones to the north of the Ross Ice Shelf (RIS) and moderate wind speeds, results in high cloud occurrence at the Ross Sea and Ross Ice Shelf. These findings align with our weather pattern observations and support the argument that synoptic weather patterns associated with high cloud occurrence will also experience higher snowfall. Despite this, further studies correlating weather patterns, cloud formation, and snowfall events are needed, as they could provide a better understanding of the region's climate.

5 Conclusions

In this study, we focused on analyzing clouds and snowfall at Scott Base, Antarctica, using data derived from lidar and radar instruments. Our goal was to investigate how likely supercooled liquid water clouds (SLWC) are to produce heavy snowfall events at this location. We demonstrated the effectiveness of lidar, specifically the Vaisala CL51 ceilometer, in detecting clouds within the Antarctic troposphere, and highlighted the radar instrument, MRR2, for its capability in detecting snowfall. While the lidar performed well in detecting clouds, it was not as effective in identifying snowfall, whereas the radar accurately detected snowfall. We presented the lidar backscatter profile and climatology, confirming the existence of SLWC within the 4 km altitude range, with backscatter values exceeding $10^{-4} \text{ m}^{-1} \text{ sr}^{-1}$. Additionally, we explored seasonal variations in cloud occurrence, discovering that austral autumn and summer exhibited higher percentages of cloud occurrence compared to winter and spring. In comparing lidar observations with MERRA2 reanalysis data, we observed that MERRA2 tends to underestimate cloud occurrence below 2 km altitude and overestimate it above 4 km. This discrepancy provides valuable insight for improving reanalysis products to better represent the Antarctic climate. We also presented the cumulative snow variations, revealing that different snow types and locations produced varying snow rates. Seasonal trends in snow accumulation were evident, with austral summer showing the highest increase in snow accumulation, which became more stagnant during austral winter. However, significant uncertainty remains in determining the appropriate Ze-Sr relationships that accurately represent snowfall in this region, due to factors such as snow shape, diameter, and mass. Finally, we compared the lidar and radar profiles and observed heavy snowfall beneath SLW cloud bases, though a more robust analysis is required to fully assess the likelihood of SLWC producing heavy snowfall. Future research could benefit from integrating lidar and radar climatology to more precisely identify regions with SLWC and other cloud contents, and to explore their correlations with precipitation events. Overall, this study has advanced our understanding of cloud and snowfall processes in Antarctica, and highlights the potential for improving climate representations through further exploration of these phenomena.

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